

Certificate verification
Connect to AIS (infrastructure access)
Verify cert_{AIS} against pk_{TA}

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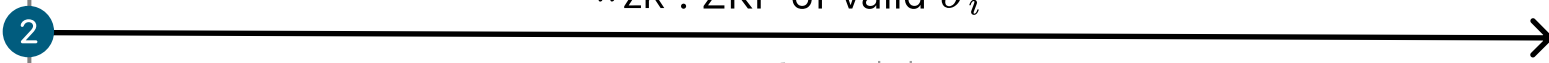
Verify OK?

fail

ABORT
Session terminated

pass

π_{ZK} : ZKP of valid σ_i



no σ_i or pk_i revealed

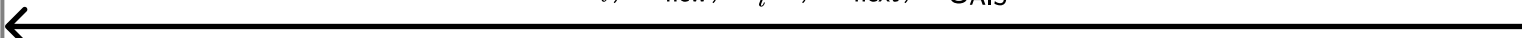
ZKP verification + AST generation
Verify π_{ZK} (credential possession)
Generate AST components:
 $\text{sid}_i \leftarrow \text{random}()$
 $t_{\text{start}}, t_{\text{end}}$: validity window (≥ 2 epochs)
 cap_i : vehicle capabilities
 $r_i \leftarrow \text{random}()$ (blinding factor)
 $\text{AST}_i = \{\text{sid}_i, t_{\text{start}}, t_{\text{end}}, \text{cap}_i, r_i\}$

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Merkle tree insertion
 $\text{leaf}_i = H_{\text{Pos}}(\text{sid}_i \parallel t_{\text{start}} \parallel t_{\text{end}} \parallel \text{cap}_i \parallel r_i)$
Insert leaf_i into active AST Merkle tree
Compute R_{new} , Merkle path π_i^{MT}
Pre-commit next epoch: R_{next} (if available)
Sign: $\text{sig}_{\text{AIS}} = \text{Sign}_{sk_{\text{AIS}}}(H(R_{\text{new}} \parallel \pi_i^{\text{MT}}))$

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$\text{AST}_i, R_{\text{new}}, \pi_i^{\text{MT}}, R_{\text{next}}, \text{sig}_{\text{AIS}}$



signed response

Offline / No connectivity needed after this point

Offline ZKP Precomputation
Verify sig_{AIS} on $(R_{\text{new}}, \pi_i^{\text{MT}})$
Generate complete Groth16 proof slots:
• Each slot committed to predicted BSM content (position, velocity, heading, timestamp)
Store complete proofs in **proof cache**
Dequeued one per BSM broadcast during $[t_{\text{start}}, t_{\text{end}}]$

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